

# Biomarker correlates of monthly savings with false-discovery-rate control

## Abstract

Background: An evaluation of a household financial coaching program. Methods: Each of the six biomarkers was correlated with monthly savings (USD) using Pearson correlation (two-sided). Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

## Introduction

We report an evaluation of a household financial coaching program. The primary question concerns participation in the financial coaching program and monthly savings (USD). The accompanying dataset (data.csv) contains one row per household-record as exported from the study database, including some duplicated exports and sentinel-coded missing values; it is described in the data dictionary.

## Methods

Participants and data. The export contains 333 records. Variables are listed in the data dictionary. Group membership is recorded in `program`.

Data cleaning and exclusions. Exact duplicate records (identical on every column) were removed, keeping the first occurrence. The value -999 denotes a missing measurement and was treated as missing. Only households aged 18 to 80 years inclusive were analysed. Records with missing savings, biomarker\_1, biomarker\_2, biomarker\_3, biomarker\_4, biomarker\_5, biomarker\_6 were excluded (complete-case analysis).

Statistical analysis. Each of the six biomarkers was correlated with monthly savings (USD) using Pearson correlation (two-sided). The six p-values were adjusted for multiple testing with the Benjamini-Hochberg false-discovery-rate procedure; associations with adjusted  $p < 0.05$  were declared significant. The strongest association is the biomarker with the largest absolute correlation. All analyses used the cleaned analysis sample; no random resampling was used.

## Results

Among 291 households, biomarker\_1 correlated with the outcome ( $r = 0.498$ , adjusted  $p < 0.001$ ), as did biomarker\_3 ( $r = -0.318$ , adjusted  $p < 0.001$ ); the adjusted  $p$  for biomarker\_5 was  $< 0.001$ . In total 3 biomarkers remained significant after adjustment; the strongest association had  $r = 0.498$ .

## Discussion

The analysis followed the pre-specified plan. Limitations include reliance on database exports and complete-case handling of missing values.